

BUSINESS INSIGHTS REPORT

Overview

This project involved cleaning, analyzing, and visualizing sales data using Excel, MySQL, and Power BI. The data was first cleaned in Excel to remove inconsistencies and prepare it for analysis. SQL was then used to query the data and identify important patterns. Finally, Power BI was used to create an interactive dashboard to visualize sales performance and business trends.

Key Findings

1. Overall Sales Performance

The company generated total sales of approximately **2.26 million**. This indicates strong business performance and a healthy customer base across different regions and product categories.

2. Regional Performance

Among all regions, the **West region** recorded the highest sales. This suggests that customers in this region contribute significantly to company revenue. The South region generated the lowest sales compared to other regions.

3. Category Performance

The **Technology** category produced the highest revenue among all product categories. Products such as technology accessories and devices appear to have strong customer demand. Furniture and Office Supplies also contributed substantially but generated lower sales than Technology.

4. Sales Trend Over Time

Sales increased steadily from **2016 to 2018**. The upward trend indicates business growth and increasing customer purchases over the years. This positive trend suggests that sales strategies and product offerings were effective during this period.

5. Customer Analysis

A small number of customers contributed a large portion of total sales. This shows that loyal and high-value customers play an important role in business success. Retaining these customers can help maintain and increase future revenue.

Recommendations

1. Increase marketing and promotional activities in high-performing regions to maximize revenue.
2. Focus on expanding the Technology category since it generates the highest sales.
3. Introduce customer loyalty programs to retain valuable customers and encourage repeat purchases.
4. Analyze successful sales strategies from the West region and apply them to lower-performing regions.
5. Continue monitoring yearly sales trends to improve forecasting and business planning.

Conclusion

The analysis provided valuable insights into sales performance, customer behavior, regional trends, and product category performance. Excel was used for data cleaning and preliminary analysis,

MySQL was used for querying and validation, and Power BI was used to create an interactive dashboard. These insights can help management make informed decisions to improve sales performance and support future business growth.

SQL QUERIES:

The screenshot shows the MySQL Workbench interface with Query 1 executed. The query is: `SELECT region, SUM(sales) AS total_sales FROM sales GROUP BY region;`. The result grid shows the following data:

region	total_sales
South	4153.6995
West	10189.2780
Central	7936.6342
East	6327.3280

The bottom panel shows the Action Output log with the following entries:

#	Time	Action	Message	Duration / Fetch
20	22:49:59	SHOW DATABASES	OK	0.000 sec
21	22:50:03	SHOW SESSION VARIABLES LIKE 'lower_case_table_names'	OK	0.000 sec
22	22:50:03	SHOW COLUMNS FROM 'salesdb' 'sales'	OK	0.000 sec
23	22:51:01	PREPARE stmt FROM 'INSERT INTO 'salesdb' 'sales' ('row_id','order_id','order_date','ship_date','ship_mode...'	OK	0.000 sec
24	22:51:02	DEALLOCATE PREPARE stmt	OK	0.000 sec
25	22:53:40	SELECT region, SUM(sales) AS total_sales FROM sales GROUP BY region LIMIT 0, 1000	4 row(s) returned	0.015 sec / 0.000 sec

The screenshot shows the MySQL Workbench interface with Query 4 executed. The query is: `SELECT category, SUM(sales) AS total_sales FROM sales GROUP BY category ORDER BY total_sales DESC;`. The result grid shows the following data:

category	total_sales
Furniture	13911.0117
Office Supplies	8322.7740
Technology	6476.1540

The bottom panel shows the Action Output log with the following entries:

#	Time	Action	Message	Duration / Fetch
21	22:50:03	SHOW SESSION VARIABLES LIKE 'lower_case_table_names'	OK	0.000 sec
22	22:50:03	SHOW COLUMNS FROM 'salesdb' 'sales'	OK	0.000 sec
23	22:51:01	PREPARE stmt FROM 'INSERT INTO 'salesdb' 'sales' ('row_id','order_id','order_date','ship_date','ship_mode...'	OK	0.000 sec
24	22:51:02	DEALLOCATE PREPARE stmt	OK	0.000 sec
25	22:53:40	SELECT region, SUM(sales) AS total_sales FROM sales GROUP BY region LIMIT 0, 1000	4 row(s) returned	0.015 sec / 0.000 sec
26	22:54:43	SELECT category, SUM(sales) AS total_sales FROM sales GROUP BY category ORDER BY total_sales DESC;	3 row(s) returned	0.031 sec / 0.000 sec

